

Lecture: Human Immunodeficiency Virus

Q1. HIV belongs to which family and genus?

- A) Picornaviridae – Enterovirus
- B) Retroviridae – Lentivirus**
- C) Flaviviridae – Hepacivirus
- D) Hepeviridae – Orthohepevirus

Q2. The first isolation of a T-lymphotropic retrovirus from a patient at risk for AIDS occurred in:

- A) 1975
- B) 1981
- C) 1983**
- D) 1990

Q3. Among HIV-1 groups, the pandemic lineage is group M, which stands for:

- A) "Middle"
- B) "Main"**
- C) "Modified"
- D) "Minor"

Q4. HIV-1 group M contains how many subtypes (letters A–U, with no E, I, or O)?

- A) 8
- B) 11
- C) 13**
- D) 18

Q5. Which antiretroviral classes is HIV-2 intrinsically resistant to?

- A) NRTIs and PIs
- B) NNRTIs and T20 (fusion inhibitor)**
- C) INSTIs and CCR5 antagonists
- D) PIs and entry inhibitors

Q6. Which recombinant form is responsible for more than 90% of HIV-1 infections in Cambodia?

- A) CRF01_AE**
- B) CRF02_AG
- C) CRF01_AB
- D) CRF11_cpx

Q7. In the HIV virion (100–120 nm), which protein forms the capsid?

- A) gp120 **envelope**
- B) p17 **matrix**
- C) p24**
- D) gp41 **envelope**

Q8. Which viral gene encodes the enzymes reverse transcriptase, integrase, and protease?

- A) gag
- B) pol**
- C) env
- D) nef

Q9. Dolutegravir, raltegravir, and elvitegravir belong to which class of antiretrovirals, and which viral enzyme do they target?

- A) NNRTIs – reverse transcriptase
- B) PIs – protease
- C) INSTIs – integrase
- D) NRTIs – reverse transcriptase

Q10. The CCR5 antagonist maraviroc blocks HIV at which step of the replication cycle?

- A) Attachment of gp120 to CD4 and co-receptor
- B) gp41-mediated membrane fusion
- C) Reverse transcription
- D) Protease-mediated maturation

Q11. Which exposure route carries the highest risk of HIV transmission?

- A) Receptive anal intercourse
- B) Mother-to-child transmission
- C) Blood transfusion
- D) Percutaneous needle-stick

Q12. Globally (UNAIDS/WHO, 2021 estimates), how many people were living with HIV?

- A) 680,000
- B) 37.7 million
- C) 1.5 million
- D) 75,000

Q13. In Cambodia, which key population has the highest estimated HIV prevalence?

- A) MSM
- B) FSW
- C) Transgender people
- D) People who inject drugs

Q14. Cambodia's progress toward the UNAIDS 95–95–95 targets is officially tracked as:

- A) 95–95–95
- B) 92–100–98
- C) 90–90–90
- D) 100–98–92

Q15. A 4th-generation ELISA differs from earlier serological tests because it simultaneously detects:

- A) HIV-1 RNA and proviral DNA
- B) Anti-HIV-1 (M and O) + anti-HIV-2 antibodies and p24 antigen
- C) Only IgG antibodies
- D) CD4 count and viral load

Q16. In the Fiebig stage, the earliest detectable biomarker after infection (Stage I, ~10–14 days) is:

- A) p24 antigen
- B) HIV antibodies by ELISA
- C) HIV RNA by nucleic acid testing
- D) Western blot bands

Q17. What is the diagnostic sensitivity and linearity range of the viral load (HIV-1 RNA) assay described in the lecture?

- A) Sensitivity 50 copies/mL; linearity 5–10⁴ copies/mL
- B) Sensitivity 20–40 copies/mL; linearity 20–10⁷ copies/mL**
- C) Sensitivity 200 copies/mL; linearity 200–10⁶ copies/mL
- D) Sensitivity 1 copy/mL; linearity 1–10⁸ copies/mL

Q18. Which regimen is the preferred first-line ART (antiretroviral treatment) in Cambodia since January 2019?

- A) AZT + 3TC + ATV/r
- B) TDF + 3TC + DTG**
- C) DRV/r + DTG
- D) TDF + FTC + EFV only

Q19. According to the clinical case on late presentation, virological failure is defined as:

- A) Any detectable viral load at 1 month
- B) Viral load > 1000 copies/mL after ≥ 3 months, confirmed**
- C) CD4 drop below 200/mm³
- D) One missed dose of ART (antiretroviral treatment)

Q20. In the newborn of an HIV-infected mother, a negative HIV DNA/RNA PCR at day 3 must be repeated after 6 weeks because:

- A) The infant PCR never becomes positive before 1 year
- B) Maternal IgG antibodies persist up to 18 months and can confuse serology**
- C) The antiviral prophylaxis interferes with PCR
- D) PCR is not recommended in newborns

Lecture: Orthomyxoviridae-Influenza virus

Q1. To which family do influenza viruses belong?

- A) Paramyxoviridae
- B) Orthomyxoviridae**
- C) Coronaviridae
- D) Pneumoviridae

Q2. How many genera are recognized within the family Orthomyxoviridae?

- A) 4
- B) 5
- C) 7**
- D) 9

Q3. Which of the following is NOT a genus of the family Orthomyxoviridae?

- A) Influenzavirus A
- B) Isavirus
- C) Thogotovirus
- D) Hepatovirus**

Q4. Influenza viruses are classified by their nucleoprotein and have which genome type?

- A) Positive-sense single-stranded RNA, non-segmented

B) Negative-sense single-stranded RNA, segmented

C) Double-stranded DNA, circular

D) Partially double-stranded DNA

Q5. Influenza A and B viruses each possess how many gene segments?

A) 6

B) 7

C) 8

D) 10

Q6. Reassortment — the exchange of viral RNA segments — has been reported:

A) Within each genus or type, but not across types

B) Across all four influenza types freely

C) Only between influenza A and B

D) Only in influenza C

Q7. How many subtypes of hemagglutinin and neuraminidase are recognized for influenza A?

A) 16 HA and 9 NA

B) 18 HA and 11 NA

C) 11 HA and 18 NA

D) 9 HA and 16 NA

Q8. Influenza B virus is classified into which two lineages?

A) H1N1 and H3N2

B) Yamagata and Victoria

C) Alpha and Beta

D) M and N

Q9. The WHO standard nomenclature system for naming influenza viruses has been in use since:

A) 1933

B) 1942

C) 1971

D) 1997

Q10. The infectious spherical form of influenza A and B virions measures:

A) 20–40 nm in diameter

B) 80–120 nm in diameter

C) 200–300 nm in diameter

D) 500 nm in diameter

Q11. Hemagglutinin on the viral surface functions primarily to:

A) Destroy sialic acid receptors

B) Bind to sialic acids on the respiratory epithelium (cell entry)

C) Bud new virions from the cell membrane

D) Block interferon signaling

Q12. Within hemagglutinin, which region is highly conserved between subtypes?

A) HA1 globular head

B) HA2 stalk region

- C) The sialic acid binding pocket
- D) The transmembrane anchor only

Q13. Neuraminidase is a sialidase that is required for which step of the viral life cycle?

- A) Attachment to the host cell
- B) Viral budding and release from the infected cell
- C) Nuclear import of viral RNA
- D) Reverse transcription

Q14. In the influenza replication cycle, the correct order of events is:

- A) Attachment → entry → release of vRNP → transcription/replication → protein production → assembly → budding
- B) Budding → attachment → replication → assembly → release
- C) Entry → assembly → attachment → budding → replication
- D) Transcription → attachment → budding → protein production → entry

Q15. Which of the following is an innate immune mediator listed in the lecture's host-response diagram?

- A) IFN- α/β
- B) Perforin only
- C) Plasma cells
- D) Antibodies

Q16. The main protective mechanisms of the adaptive immune response to influenza include:

- A) Antibody-mediated virus neutralization and CD8⁺ T-cell cytotoxicity
- B) Only natural killer cell activity
- C) Mucus hypersecretion
- D) Blockade of neuraminidase by complement alone

Q17. Match the definition: "an outbreak of disease that attacks many people at about the same time and may spread through one or several communities" is:

- A) Endemic
- B) Epidemic
- C) Pandemic
- D) Sporadic

Q18. In temperate climates, seasonal influenza epidemics occur mainly:

- A) During winter (cold months)
- B) During summer
- C) Throughout the year equally
- D) Only during the rainy season

Q19. Annual seasonal influenza epidemics are estimated to cause:

- A) 100,000–200,000 deaths
- B) 3–5 million cases of severe illness and about 290,000–650,000 deaths
- C) 1–2 million cases of severe illness and 50,000 deaths
- D) 50 million cases of severe illness

Q20. Which group is **NOT** listed among those at greater risk of severe influenza or complications?

- A) Pregnant women

- B) Children under 59 months
- C) Healthy adults aged 20–40 years
- D) Healthcare workers

Q21. The incubation period of seasonal influenza is:

- A) 1–4 days (usually 2)
- B) 7–14 days
- C) 14–21 days
- D) Less than 12 hours

Q22. Which symptom is characteristic of seasonal influenza and may last 2 or more weeks?

- A) Runny nose only
- B) Dry cough
- C) Conjunctivitis
- D) Jaundice

Q23. For laboratory confirmation of influenza, which method is considered the gold standard?

- A) Rapid influenza diagnostic tests
- B) Virus isolation (cell culture or eggs)
- C) Direct antigen detection only
- D) Clinical examination alone

Q24. Compared with RT-PCR, rapid influenza diagnostic tests:

- A) Have higher sensitivity
- B) Have lower sensitivity and can be unreliable
- C) Detect all influenza C subtypes
- D) Are the gold standard

Q25. In the history of influenza vaccines, the first bivalent vaccine appeared in:

- A) 1933
- B) 1942
- C) 1968
- D) 1978

Q26. Which statement correctly describes the adjuvant MF59?

- A) A mineral salt (alum) used in more than 80% of licensed vaccines
- B) An oil-in-water emulsion approved for influenza, introduced in 1997
- C) A liposome (virosome) approved for influenza and hepatitis A
- D) An alum-adsorbed TLR4 agonist used for HPV and HBV vaccines

Q27. Why are adamantanes (e.g., amantadine) no longer recommended for influenza?

- A) They target neuraminidase and are too toxic
- B) Resistance is approximately 99%
- C) They are ineffective in children
- D) They cannot cross the blood–brain barrier

Q28. Neuraminidase inhibitors prevent viral budding by:

- A) Blocking the M2 protein channel
- B) Blocking the neuraminidase enzyme
- C) Inhibiting RNA polymerase

D) Blocking HA binding to sialic acids

Q29. The first human influenza A(H5N1) cases occurred in:

- A) Hong Kong in 1997 and China in 2003
- B) Cambodia in 2004
- C) Vietnam in 2005
- D) The USA in 2009

Q30. In Cambodia, the cumulative A(H5N1) cases at the time of the lecture was:

- A) 56 human cases with 37 deaths (CFR 66%)
- B) 100 human cases with 2 deaths
- C) 20 human cases with no deaths
- D) 859 human cases with 453 deaths

Lecture: Arboviruses

Q1. What does the term "arbovirus" mean?

- A) Airborne-borne virus
- B) Arthropod-borne virus
- C) Animal-borne virus
- D) Avian-borne virus

Q2. Which of the following families is NOT listed among the arbovirus families in the lecture?

- A) Flaviviridae
- B) Alphaviridae
- C) Bunyaviridae
- D) Hepadnaviridae

Q3. Which arbovirus is transmitted to humans primarily by *Culex* mosquitoes?

- A) Dengue virus
- B) Zika virus
- C) Japanese encephalitis virus
- D) Chikungunya virus

Q4. Among these arboviruses, which one is transmitted by ticks?

- A) Rift Valley Fever virus
- B) Crimean-Congo hemorrhagic fever virus
- C) Banna virus
- D) Equine encephalitis virus

Q5. Dengue virus belongs to which genus, and what type of genome does it carry?

- A) Alphavirus – negative-sense RNA
- B) Flavivirus – enveloped, positive-sense single-stranded RNA (+ssRNA)
- C) Orthomyxovirus – segmented RNA
- D) Hepacivirus – double-stranded DNA

Q6. During flavivirus polyprotein processing, the polyprotein is cleaved into how many structural and non-structural proteins?

- A) 2 structural and 5 non-structural

C= Capsid
E= Envelope

B) 3 structural (C, prM, E) and 7 non-structural (NS1–NS5)

C) 4 structural and 8 non-structural

D) 5 structural and 3 non-structural

Q7. Which viral enzyme replicates the flavivirus +ssRNA genome in the cytoplasm?

A) NS3 protease

B) NS5 RNA-dependent RNA polymerase

C) Viral reverse transcriptase

D) Furin

Q8. During flavivirus maturation, which host enzyme cleaves the prM protein into the mature M protein?

A) Furin

B) Trypsin

C) Caspase

D) Neuraminidase

Q9. Where does the entire flavivirus life cycle take place?

A) In the host cell nucleus

B) Entirely in the host cell cytoplasm

C) In the mitochondria

D) In the lysosome only

Q10. How many serotypes and genotypes does Zika virus possess?

A) 4 serotypes and 2 genotypes

B) 1 serotype and 2 genotypes Genotypes: African and Asian

C) 2 serotypes and 4 genotypes

D) 1 serotype and 1 genotype

Q11. Which complications are associated with Zika virus infection?

A) Congenital brain abnormalities including microcephaly, and Guillain-Barré syndrome

B) Chronic hepatitis and cirrhosis

Zika virus linked to neurological disorders

C) Myocarditis and rheumatic fever

D) Acute flaccid paralysis only

Q12. According to the lecture, how are ZIKV concentrations ordered among body fluids?

A) Plasma < saliva < urine < semen < whole blood

B) Whole blood < plasma < urine < saliva < semen

C) Saliva < plasma < semen < urine < whole blood

D) Urine < plasma < saliva < whole blood < semen

Q13. The first large ZIKV outbreak in humans occurred in 2007 on the island of Yap. How many human ZIKV cases had been documented worldwide before this outbreak?

A) 0

B) 14

C) 250

D) 5,000

Q14. In Cambodia, the first ZIKV report found one RT-PCR–positive case from which province, among ~10,000 participants with acute fever?

- A) Kandal
- B) Siem Reap
- C) Kampong Speu
- D) Battambang

Q15. Which statement about Japanese encephalitis virus is correct?

- A) Humans are the reservoir and transmit JEV person-to-person
- B) Domestic pigs and wild birds are reservoirs; humans are incidental dead-end hosts
- C) Cattle are the only reservoir
- D) JEV is transmitted by *Aedes aegypti* in urban areas

Q16. When symptomatic, Japanese encephalitis has a case-fatality rate of up to:

- A) 5%
- B) 15%
- C) 30%
- D) 80%

Q17. Arboviruses are a diverse group of viruses that belong to several different families. Which of the following viral families is commonly associated with arboviruses?

- A) Herpesviridae
- B) Retroviridae
- C) Flaviviridae and Togaviridae
- D) Adenoviridae

Q18. Which JEV genotype was detected in Cambodia (from encephalitic children, pigs, and mosquitoes)?

- A) Genotype 3
- B) Genotype 5
- C) Genotype 1 (subtypes 1a and 1b), with lineages close to Thai, Vietnamese, and Lao strains
- D) Genotype 2 only

Q19. Chikungunya virus is classified into which three genotypes?

- A) Asian, West African, and ECSA (with an Indian Ocean lineage)
- B) African, European, and American
- C) Clade I, clade II, and clade III
- D) DENV-1, DENV-2, and DENV-3

Q20. What is the biological consequence of the CHIKV E1-A226V mutation?

- A) Loss of infectivity for *Aedes* mosquitoes
- B) Increased infectivity and transmissibility by *Aedes albopictus*
- C) Resistance to antiviral drugs
- D) Conversion to a tick-borne virus

Q21. In the 2012 Kampong Speu CHIKV outbreak in Cambodia, what proportion of febrile patients tested positive (188/425)?

- A) 12.3%
- B) 24.5%
- C) 44.2%
- D) 65.7%

Q22. How many dengue virus serotypes exist, and how is each serotype further classified?

- A) 2 serotypes, classified by vector species
- B) 4 serotypes (DENV-1 to DENV-4), each classified into several geographic genotypes
- C) 5 serotypes, each classified by cell tropism
- D) 4 serotypes with no further classification

Q23. Which figures describe the global burden of dengue according to the lecture?

- A) 390 million cases/year; 3.9 billion people in 128 countries at risk
- B) 39 million cases/year; 1 billion at risk
- C) 390,000 cases/year; 500 million at risk
- D) 3.9 million cases/year; 200 million at risk

Q24. In Cambodia, three large dengue outbreaks occurred in 2007, 2012, and 2019. Which serotypes were dominant respectively?

- A) D1, D2, D3
- B) D3, D1, D1
- C) D4, D3, D2
- D) D2, D4, D3

Q25. Which statement about NS1 antigen detection for dengue diagnosis is correct?

- A) It is positive only after antibody seroconversion
- B) It appears early, correlates with viremia, and remains detectable in blood longer than viremia
- C) It cannot be detected by rapid diagnostic tests
- D) It distinguishes dengue from Japanese encephalitis infections

Q26. Which of the following diseases is caused by an arbovirus?

- A) Common cold
- B) Polio
- C) Dengue Fever
- D) Influenza

Q27. Besides specific vaccines for some arboviruses, what personal prevention measure is recommended to reduce contact with vectors?

- A) Eating specific foods
- B) Avoiding all contact with other people
- C) Using insect repellents and wearing protective clothing
- D) Taking daily antiviral medication

Q28. What is a common method used for the laboratory diagnosis of arboviral infections?

- A) X-ray imaging
- B) Blood culture for bacteria
- C) Serological tests like ELISA and molecular tests like RT-PCR
- D) Fecal examination

Q29. Which of these arboviruses can also be transmitted sexually, in addition to arthropod bites?

- A) Dengue virus
- B) West Nile virus
- C) Zika virus
- D) Yellow Fever virus

Q30. Many arboviruses cause a range of symptoms, from mild febrile illness to severe neuroinvasive disease. Which of these is a severe neurological manifestation caused by some arboviruses?

- A) chronic fatigue syndrome
- B) Meningitis and Encephalitis**
- C) Multiple sclerosis
- D) Parkinson's disease

Q31. Which of the following virus is caused joint pain for long time?

- A) Chikungunya virus**
- B) Influenza
- C) Zika virus
- D) Coronavirus

Q32. Which of the following assay is used to detect total immunoglobulins of Arboviruses?

- A) Inhibition of Hemagglutination Assay**
 - B) Virus isolation
 - C) RT-PCR
 - D) NS1 antigen detection
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Lecture: Ebola virus

Q1. How many genera are recognized in the family Filoviridae, and what are they?

- A) 2 — Marburgvirus and Ebolavirus
- B) 3 — Cuevavirus, Marburgvirus, and Ebolavirus**
- C) 4 — Cuevavirus, Ebolavirus, Restonvirus, and Marburgvirus
- D) 5 — one per Ebolavirus species

Q2. As classified by WHO, filoviruses are Risk Group:

- A) 2
- B) 3
- C) 4**
- D) 1

Q3. Since its discovery in 1976, Ebola disease has mostly occurred in:

- A) Southern Asia
- B) Sub-Saharan Africa**
- C) South America
- D) Western Europe

Q4. The first Ebola cases were detected in 1976 in which two countries?

- A) Guinea and Liberia
- B) Uganda and Kenya
- C) Democratic Republic of Congo and Sudan**
- D) Gabon and Cameroon

Q5. In 1967, importation of infected green monkeys (*Cercopithecus aethiops*) from Uganda caused Marburg virus disease outbreaks in:

- A) Germany and Yugoslavia, causing 32 fatalities**
- B) France and Spain, causing 5 fatalities

- C) The USA and Canada
- D) Brazil and Argentina

Q6. The Ebola virus genome is characterized as:

- A) Double-stranded DNA, circular, 190 kb
- B) Single-stranded, helical, negative-sense RNA, ~19 kb, enveloped
- C) Positive-sense RNA, segmented, 8 genes
- D) Enveloped DNA with reverse transcriptase

Q7. How many distinct species of Ebolavirus?

- A) 3
- B) 4
- C) 5 Bundibugyo, Tai Forest, Reston, Sudan, Zaire
- D) 7

Q8. Which Ebolavirus species is the one abbreviated EBOV?

- A) Sudan ebolavirus
- B) Zaire ebolavirus
- C) Reston ebolavirus
- D) Bundibugyo ebolavirus

Q9. What is the natural reservoir of Ebola virus, and how is it transmitted?

- A) Bats; by direct contact with blood and body fluids of infected persons or animals, or contaminated objects
- B) Mosquitoes; by bite
- C) Pigs; by aerosol in crowded settings
- D) Ticks; by tick bite

Q10. According to the lecture, men who have recovered from Ebola can still transmit the virus through semen for up to:

- A) 7 days
- B) 7 weeks
- C) 6 months
- D) 2 years

Q11. The incubation period of Ebola virus disease is:

- A) 1–3 days
- B) 2–21 days
- C) 21–45 days
- D) 1–2 weeks in all cases

Q12. Which of the following is characteristic of Stage I (unspecific) Ebola disease?

- A) Bleeding from eyes, ears, and nose
- B) Extreme asthenia, headaches, myalgia, arthralgia, conjunctivitis, and a rash over the body except the face
- C) Unilateral orchitis and hearing loss
- D) Jaundice and splenomegaly

Q13. Which statement about Stage II (specific) Ebola disease is correct?

A) Internal and external bleeding occurs, and mortality in this phase is 50–90%

B) Recovery is expected in > 90% without treatment

C) Bleeding is limited to the gastrointestinal tract

D) Neurological sequelae occur in most survivors

Q14. Which laboratory diagnostic tests are recommended within a few days after Ebola symptoms begin?

A) Antigen-capture ELISA, IgM ELISA, PCR, and virus isolation

B) IgG ELISA only

C) Immunohistochemistry on skin biopsy

D) Antibody avidity testing

Q15. According to the lecture, which statement about Ebola treatment and prevention is correct?

A) Interferons are effective; a licensed vaccine is available

B) No antiviral drugs are available; interferons are not effective; and hospital isolation facilities are the most important preventive measure

C) Ribavirin is the first-line antiviral

D) Vaccination of contacts is the primary control measure

Q16. Ebola virus, belong to which viral family?

A) Flaviviridae

B) Herpesviridae

C) Picornaviridae

D) Filoviridae

Q17. Which of the following animals is considered a primary reservoir for Ebola virus?

A) Birds

B) Rodents

C) Bats

D) Dogs

Lecture: Coronaviruses

Q1. How many subfamilies does the family Coronaviridae contain, and what are they?

A) 2 — Letovirinae and Torovirinae

B) 3 — Letovirinae, Torovirinae, and Orthocoronavirinae

C) 1 — Orthocoronavirinae only

D) 4 — including Paramyxovirinae

Q2. Coronaviruses are characterized as:

A) Enveloped, negative-sense RNA viruses with segmented genomes

B) Enveloped, positive-sense single-stranded RNA (+ssRNA) viruses with unsegmented genomes of 26–32 kb

C) Non-enveloped DNA viruses

D) Enveloped dsRNA viruses

Q3. Which glycoprotein of the coronavirus envelope is responsible for receptor binding, cell fusion, and is the major antigen?

A) M protein associated with nucleocapsid

B) E protein small, coat protein associated

C) S (spike) protein

D) HE protein hemagglutinin-esterase protein (in group 2)

Q4. The hemagglutinin-esterase glycoprotein surface projection (5–7 nm) is found in:

A) All coronaviruses

B) Group 1 alphacoronaviruses only

C) Group 2 betacoronaviruses

D) Gammacoronaviruses only

Q5. Which receptor does the human coronavirus 229E use for cell entry?

A) ACE2

B) Aminopeptidase N

C) CEACAM

D) TMPRSS2

Q6. SARS-CoV and NL63 both use which cellular receptor?

A) CD13

B) N-acetyl-9-O-acetylneuraminic acid

C) ACE2 (angiotensin-converting enzyme 2)

D) CD4

Q7. The OC43 coronavirus attaches to cells via which receptor?

A) ACE2

B) APN

C) N-acetyl-9-O-acetylneuraminic acid or CEACAM

D) Integrin $\alpha v \beta 3$

Q8. By cryo-electron tomography, the coronavirus envelope is unusually thick. Its measured thickness is:

A) 2 nm

B) 4 nm

C) 7.8 ± 0.7 nm

D) 20 nm

Q9. The viral genome of coronaviruses is associated with which protein to form the nucleocapsid?

A) Spike (S)

B) Membrane (M)

C) Phosphoprotein nucleocapsid protein (N)

D) Envelope (E)

Q10. The coronavirus genome has which terminal features?

A) 5' cap and 3' poly(A) tract

B) Covalently linked terminal protein at both ends

C) 5' poly(A) and 3' cap

D) Hairpin loops only

Q11. After entry, the coronavirus genome is translated into two polyproteins (pp1a and pp1ab) from ORF1a and ORF1b, which are cleaved into:

A) 5 structural proteins

B) 16 non-structural proteins forming the replication–transcription complex

C) 3 glycoproteins

D) 12 capsid proteins

Q12. Which structures are characteristic of the coronavirus replication organelles?

A) Double-membrane vesicles, convoluted membranes, and double-membrane spherules

B) Negri bodies

C) Guarnieri bodies

D) Nuclear inclusion bodies

Q13. Which virus became the 7th human coronavirus at the end of 2019?

A) MERS-CoV

B) SARS-CoV

C) SARS-CoV-2 (2019-nCoV), causing COVID-19 in Wuhan, China

D) HKU1

Q14. Among the seven human coronaviruses, which four cause mild, flu-like respiratory infections?

A) SARS-CoV, MERS-CoV, SARS-CoV-2, and HKU1

B) 229E, NL63, OC43, and HKU1

C) 229E, OC43, SARS-CoV-1, and NL63

D) MERS-CoV, SARS-CoV-1, HKU1, and OC43

Q15. During the 2002–2004 SARS-CoV outbreak, the mortality rate was approximately:

A) 1%

B) 10% (spread across 29 countries)

C) 35%

D) 50%

Q16. MERS-CoV emerged in 2012 in Saudi Arabia. As of January 2020, its overall mortality rate among confirmed cases was approximately:

A) 2.5%

B) 10%

C) 35%

D) 66%

Q17. According to the lecture, COVID-19 is transmitted by:

A) Droplets ($> 5 \mu\text{m}$) and aerosols ($< 5 \mu\text{m}$) with respiratory secretions, plus contact with contaminated surfaces

B) Blood transfusion only

C) Faecal–oral route only

D) Vector-borne transmission

Q18. What is the estimated severity distribution of COVID-19?

A) 50% mild, 40% severe, 10% critical

B) 80% mild to moderate, 15% severe, 5% critical

C) 95% critical, 5% mild

D) 60% asymptomatic, 40% severe

Q19. What proportion of COVID-19 deaths were associated with underlying conditions, and what is the incubation period?

- A) 25% of deaths; incubation 1–2 days
- B) 75% of deaths; incubation 5–6 days, up to 14 days**
- C) 50% of deaths; incubation 14–21 days
- D) 90% of deaths; incubation 2–7 days

Q20. Why is serology testing not recommended for COVID-19 diagnosis in Cambodia?

- A) Antibodies appear late (7–14 days after symptom onset), and the Ministry of Health uses only tests that detect the virus**
- B) The tests are too expensive
- C) Antibodies never develop after infection
- D) PCR cannot detect SARS-CoV-2 RNA

Q21. The outbreak of coronavirus that causes severe acute respiratory syndrome was first identified in 2003.

- A) SARS-CoV-2
- B) SARS-CoV-1**
- C) MERS-CoV
- D) Covid-19

Q22. When was the first case of the COVID-19 infection reported in the China?

- A) 2020
- B) 2019**
- C) 2018
- D) 2017

Q23. Which of the following virus may not show flu-like symptoms in humans?

- A) Coronavirus
- B) Rhinovirus
- C) Rotavirus**
- D) Influenza virus

Q24. Which of the following statement is incorrect about the important characteristics/properties of Coronavirus?

- A) Non-Enveloped RNA virus**
- B) Enveloped RNA virus
- C) Distributed around the world
- D) Transmitted from person to person

Q25. The middle east respiratory syndrome (MERS) is a type of coronavirus, which of the following year the first case of the MERS virus was reported?

- A) 2003
- B) 2020
- C) 2012**
- D) 2019

Q26. The most likely source of the MERS-CoV to humans is

- A) Dogs

- B) Fishes
- C) Camels**
- D) Birds

Q27. What is the reservoir origin of Coronavirus?

- A) Birds
- B) Pigs
- C) Pangolins
- D) Bats**

Q28. All of the following are the common symptoms caused by the COVID-19 infection, Except?

- A) Loss of taste or smell
- B) Difficulty in breathing
- C) Sore throat and cough
- D) Brown spots around the face**

Q29. The outbreak of coronavirus (Covid-19) pandemic began.

- A) In the Amazon jungle in South America
- B) We have no way of knowing where the outbreak began
- C) In Wuhan, China in late 2019**
- D) In the United States, in Kirkland, WA, late February 2020

Q30. The best and most accurate way to diagnose Covid-19 is

- A) There is no best or accurate way to diagnose Covid
 - B) A qualitative Covid polymerase chain reaction (PCR) or molecular test**
 - C) Identifying symptoms, such as fever, cough, or sore throat
 - D) Serology test
-

Lecture: Poliovirus

Q1. To which family and genus does poliovirus belong?

- A) Orthomyxoviridae – Influenzavirus
- B) Picornaviridae – Enterovirus C**
- C) Rhabdoviridae – Lyssavirus
- D) Flaviviridae – Flavivirus

Q2. How many serotypes of poliovirus exist?

- A) 1
- B) 2
- C) 3 (Poliovirus 1, 2, and 3)**
- D) 4

Q3. Enteroviruses are a genus of viruses belonging to which family?

- A) Coronaviridae
- B) Flaviviridae
- C) Picornaviridae**
- D) Adenoviridae

Q4. How is poliovirus transmitted, and what is its only natural reservoir?

- A) Respiratory droplets; bats
- B) Oro-faecal or oro-oral route; humans only**
- C) Sexual contact; primates
- D) Vector-borne; rodents

Q5. Poliovirus is inactivated by all of the following EXCEPT:

- A) Heating at 55 °C for 30 minutes
- B) Chlorine at a concentration of 0.1 ppm
- C) Bleach and UV radiation

resistant to **D) Stomach acidity**

Q6. What proportion of poliovirus infections are inapparent (asymptomatic)?

- A) 5%
- B) 50%
- C) 75%
- D) 95%**

Q7. Which clinical form of polio is described as a "minor illness" with fever, headache, sore throat, listlessness, and a normal neurological examination?

- A) Paralytic poliomyelitis
- B) Abortive poliomyelitis**
- C) Non-paralytic (aseptic meningitis)
- D) Polio-encephalitis

Q8. Non-paralytic poliomyelitis (aseptic meningitis) is characterized by:

- A) Asymmetric flaccid paralysis
- B) Signs of meningeal irritation: severe headache, vomiting, cutaneous hyperesthesia, neck stiffness — with full recovery**
- C) Spastic paralysis in infants
- D) Respiratory center paralysis

Q9. Which of the following is the primary route of transmission for Enteroviruses?

- A) Airborne droplets
- B) Sexual contact
- C) Fecal-oral route**
- D) Vector-borne (e.g., mosquito bites)

Q10. Which poliovirus type caused the disease in 85% of cases?

- A) Type 1**
- B) Type 2
- C) Type 3
- D) Type 4

Q11. Which form of paralytic polio is the most severe, causing 75% of deaths?

- A) Spinal paralytic polio
- B) Bulbar (brain stem) paralytic polio**
- C) Non-paralytic polio
- D) Abortive polio

Q12. For laboratory diagnosis, poliovirus is most reliably isolated from:

- A) CSF during the first week only
- B) The throat during the first week of illness and from stool for several weeks**
- C) Urine for up to 6 months
- D) Blood culture at day 14

Q13. Jonas Salk's inactivated polio vaccine, announced in April 1955:

- A) Is a live attenuated trivalent vaccine given orally
- B) Produces immunity in >90% after 2 doses and >99% after 3 doses**
- C) Is available only in Europe
- D) Requires only 1 dose for lifelong immunity

Q14. Albert Sabin's oral polio vaccine is characterized by all of the following EXCEPT:

- A) It is a live attenuated vaccine containing Sabin types 1, 2, and 3
- B) It produces immunity in >95% after 3 doses
- C) It can mutate and regain its neurovirulence
- D) It does not induce local intestinal immunity**

Q15. Hand, foot, and mouth disease is a common childhood illness primarily caused by which group of viruses?

- A) Rhinoviruses
- B) Rotaviruses
- C) Enteroviruses**
- D) Noroviruses

Q16. While many Enterovirus infections are mild, more severe cases involving the central nervous system can be difficult to diagnose without specific laboratory tests. Which method is most crucial for a definitive diagnosis of Enterovirus infections?

- A) Physical examination only
- B) Patient symptom history alone
- C) Nucleic acid amplification tests**
- D) X-ray imaging

Q17. Poliovirus infection primarily targets which part of the human body to cause paralytic poliomyelitis?

- A) Lungs
- B) Heart
- C) Central Nervous System**
- D) Liver

Q18. Which of the following is a common, severe outcome of poliovirus infection, although it only occurs in a small percentage of cases?

- A) Mild rash
- B) Paralysis**
- C) Common cold symptoms
- D) Stomachache

Q19. Which type of poliovirus vaccine replicates in the gut, elicits better primary intestinal immunity, and more effectively prevents poliovirus transmission?

- A) Inactivated Poliovirus Vaccine

B) Oral Poliovirus Vaccine

C) Subunit vaccine

D) Recombinant vaccine

Q20. What is Post-Polio Syndrome?

A) An acute phase of poliovirus infection

B) A condition that can affect polio survivors' decades after initial recovery, causing new muscle weakness, pain, and fatigue

C) A mild allergic reaction to the polio vaccine

D) A respiratory illness caused by poliovirus

Lecture: Lyssavirus and Rabies

Q1. The rabies virus belongs to which family and genus?

A) Picornaviridae – Enterovirus

B) Rhabdoviridae – Lyssavirus

C) Paramyxoviridae – Morbillivirus

D) Coronaviridae – Betacoronavirus

Q2. The rabies virion is described as:

A) Spherical, enveloped, positive-sense RNA

B) Bullet-shaped, enveloped, with a negative-sense single-stranded RNA genome

C) Icosahedral, non-enveloped, double-stranded DNA

D) Filamentous, segmented RNA

Q3. Where does rabies virus replicate, and why?

A) In the nucleus, because it uses host RNA polymerase

B) Entirely in the host cell cytoplasm, because the virus carries its own copying machinery (host cells cannot directly read negative-sense RNA)

C) In mitochondria, using mitochondrial polymerase

D) In the Golgi apparatus only

Q4. How many major species (genotypes) of lyssavirus are described, and which one is NOT associated with human rabies?

A) 5 species; Duvenhage virus

B) 7 species; Lagos bat virus (genotype 2)

C) 7 species; Mokola virus

D) 4 species; Australian bat lyssavirus

Q5. Which animals are the principal reservoir hosts for many lyssaviruses?

A) Dogs only

B) Cats and foxes

C) Bats (order Chiroptera)

D) Rodents

Q6. During attachment, the viral glycoprotein (G) spikes bind to receptors found abundantly on nerve cells. Which receptors are listed?

A) ACE2 and CD4

B) Nicotinic acetylcholine receptor (nAChR), neural cell adhesion molecule, and p75 neurotrophin receptor

C) Aminopeptidase N and CEACAM

D) Integrins only

Q7. Which of the following is listed as a route of rabies transmission to humans?

A) Bite and scratches via saliva of an infected animal

B) Blood transfusion

C) Faecal–oral contamination of food

D) Sexual contact

Q8. The incubation period for rabies in humans can be highly variable, but it commonly ranges from:

A) A few hours to a few days

B) 1 to 2 weeks

C) 30 to 90 days (1-3 months)

D) Several years (e.g., 5-10 years)

Q9. Which statement correctly describes the two clinical forms of human rabies?

A) Paralytic form (20%) and furious form (hydrophobia, aerophobia, hyper-excitation)

B) Hepatic form and respiratory form

C) Paralytic form (80%) and furious form (20%)

D) Cutaneous form and meningeal form

Q10. Signs of the furious form of rabies include all of the following EXCEPT:

A) Hydrophobia

B) Hyperesthesia at the pharynx and larynx

C) Aerophobia

D) Prolonged remission with full recovery

Q11. For laboratory diagnosis, the most frequently used test, performed on brain specimens, is:

A) RT-PCR on saliva

B) Direct fluorescent antibody test (dFA) — fast (2–3 hours), sensitive and specific, used for post-mortem diagnosis

C) Antigen ELISA on urine

D) Serology for IgG

Q12. Which technique allows ante-mortem diagnosis of rabies by detecting viral RNA?

A) Direct fluorescent antibody test (dFA) on brain smear

B) Reverse transcriptase-polymerase chain reaction on brain, saliva, CSF, skin biopsy, or hair follicle

C) Virus isolation in cell culture

D) Tzanck smear

Q13. Louis Pasteur's 1885 nerve tissue vaccine was characterized by:

A) Recombinant protein technology

B) Use of desiccated spinal cord of rabbit, with a risk of allergic reaction and 13 injections at the belly

C) Three intradermal sessions

D) Cell-culture purification

Q14. In modern rabies post-exposure prophylaxis, the correct approach is:

- A) Wound washing with soap and water for ~10 minutes, antiseptic application, then vaccine (IM 4 sessions or ID 3 sessions), plus rabies immunoglobulin for WHO Category 3 wounds
- B) Immediate wound suturing before any washing
- C) Vaccine only, without wound care
- D) Rabies immunoglobulin for all wounds regardless of severity

Q15. Regarding rabies in Cambodia, which statement is correct?

- A) About 600,000 dog bites occur per year; 80% are severe; 60% of victims are children under 17; only about 5% consulted a rabies vaccination center
- B) Cambodia has no rabies cases
- C) Dogs are tested yearly with 100% positivity
- D) Most bites occur in adults over 40 years of age

Q16. Which of these clinical signs is most characteristic of human rabies infection?

- A) Rash on the skin
- B) Severe joint pain
- C) Hydrophobia (fear of water)
- D) Jaundice

Q17. Which statement about the prognosis of rabies once clinical symptoms appear is generally true?

- A) It is easily treatable with antiviral medication.
- B) Recovery is common with supportive care.
- C) 100% fatal.
- D) Patients typically make a full recovery if diagnosed early.

Lecture: Papillomaviridae

Q1. HPV particles are characterized as:

- A) Large, enveloped, negative-sense RNA viruses
- B) Small, non-enveloped, icosahedral viruses measuring 52–55 nm, containing double-stranded DNA
- C) Bullet-shaped, enveloped DNA viruses
- D) Non-enveloped RNA viruses with segmented genomes

Q2. The HPV capsid is composed of:

- A) 72 pentameric capsomers formed by two structural proteins, L1 (55 kDa) and L2 (70 kDa)
- B) 20 hexameric capsomers of a single protein
- C) 12 pentamers of E1 and E2
- D) 180 capsomers of the E6 protein

Q3. Which statement about the L1 capsid protein is correct?

- A) L1 makes up 80% of viral proteins and its expression alone can produce virus-like particles
- B) L1 is a non-structural protein involved in replication
- C) L1 is variably present and not surface-exposed
- D) L1 is encoded by the early (E) region

Q4. The ~8000 bp HPV genome is divided into which three functional regions?

- A) 5'UTR, capsid, and 3'UTR
- B) Early (E), late (L), and long control region
- C) gag, pol, and env

D) Alpha, beta, and gamma

Q5. Which HPV oncoproteins are involved in immune evasion by downregulating MHC class I and disrupting the interferon pathway?

A) L1 and L2

B) E5, E6, and E7

C) E1 and E2

D) E4 only

Q6. HPV types 6 and 11 are most strongly associated with:

A) Cervical cancer

B) Anogenital warts and recurrent respiratory papillomatosis

C) Cutaneous warts on hands and feet

D) Oropharyngeal cancer only

Q7. Which HPV types are strongly linked to cancers of the cervix, anogenital areas, and oropharynx?

A) HPV 6 and 11

B) HPV 16 and 18

C) HPV 1 and 2

D) HPV 4 and 27

Q8. Papillomaviruses primarily infect which type of cells?

A) Neurons

B) Liver cells

C) Epithelial cells

D) Blood cells

Q9. The primary mode of transmission for Human Papillomavirus is:

A) Airborne droplets

B) Contaminated food and water

C) Sexual contact

D) Vector-borne

Q10. Which HPV types cause cutaneous warts on hands and feet, transmitting via skin contact or fomites?

A) 16, 18, 31, and 33

B) 6, 11, 42, and 43

C) 1, 2, 4, 27, and 57

D) 51, 52, 56, and 58

Q11. How does HPV avoid immune detection?

A) By not lysing cells and causing no viremia, by downregulating MHC class I, and by disrupting the interferon pathway

B) By replicating in the nucleus and integrating into the genome

C) By causing massive cell lysis

D) By producing immunosuppressive toxins

Q12. Which laboratory technique can differentiate "low-risk" (6, 11, 42, 43, 44) from "high-risk" (16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 68) HPV genotype groups?

A) Hybrid Capture HPV DNA Test 2 (hc2)

B) Viral culture on HeLa cells

C) Peripheral blood smear

D) Direct fluorescent antibody test

- Q13.** Regarding the HPV vaccine, which statement is correct?
- A) It is a live attenuated vaccine given intradermally
 - B) A 9-valent vaccine is licensed; it uses the L1 major capsid protein produced by recombinant technology as virus-like particles, is given intramuscularly, and routine vaccination is recommended at age 11–12 for both females and males
 - C) It contains the whole inactivated virus and is licensed for people over 45
 - D) It is a DNA vaccine encoding E6 and E7
- Q14.** According to the document, what proportion of HPV infections resolve spontaneously and within what time?
- A) 90% resolve spontaneously within two years
 - B) 50% resolve within 6 months
 - C) 10% resolve within 5 years
 - D) No infections resolve spontaneously
- Q15.** Which factors are stated to "trigger" HPV-associated cancer, since HPV itself does not directly cause cancer?
- A) Smoking, exposure to UV and other carcinogenic chemicals, folate deficiency, immunosuppression, and pregnancy
 - B) Excessive exercise and dietary fiber
 - C) Cold climate and altitude
 - D) Antibiotic overuse only
- Q16.** What is considered the most efficient approach to reducing the burden of HPV-related diseases?
- A) Regular antibiotic treatment
 - B) Surgical removal of all lesions
 - C) Vaccination against HPV
 - D) Dietary changes
- Q17.** Approximately how many distinct HPV genotypes have been identified?
- A) Less than 50
 - B) Between 50 and 100
 - C) Over 100
 - D) Exactly 500
-

Lecture: Poxvirus

- Q1.** The family *Poxviridae* is characterized by which genome type and structure?
- A) Small, non-enveloped, negative-sense RNA viruses
 - B) Large, enveloped, double-stranded DNA viruses infecting vertebrates and invertebrates
 - C) Enveloped, positive-sense RNA viruses with segmented genomes
 - D) Non-enveloped, circular DNA viruses
- Q2.** The poxvirus genome ranges from 130–360 kb and usually encodes:
- A) 5–10 genes
 - B) 70–200 genes
 - C) More than 150 genes
 - D) Exactly 8 genes
- Q3.** Which subfamily of *Poxviridae* includes all human poxviruses?
- A) Entomopoxvirinae insects
 - B) Chordopoxvirinae

- C) Orthopoxvirinae
- D) Molluscipoxvirinae

Q4. What is the unique feature that distinguishes poxviruses from other DNA viruses such as herpesviruses and adenoviruses?

- A) They replicate entirely in the cytoplasm
- B) They replicate in the nucleus
- C) They integrate their genome into the host chromosome
- D) They are non-enveloped

Q5. Which virus is the only poxvirus whose natural host is only humans, causing a benign skin infection with small, firm, pearly papules?

- A) Cowpox virus
- B) Vaccinia virus
- C) Molluscum contagiosum virus
- D) Camelpox virus

Q6. The monkeypox virus particle measures approximately:

- A) 20–40 nm in diameter
- B) 200–450 nm long × 160–260 nm wide, with a genome of ~197 kb
- C) 55–65 nm, with a genome of 9.6 kb
- D) 100–120 nm, with a genome of 3,200 bp

Q7. Which Orthopoxvirus member has the only natural host of humans and caused the disease eradicated in 1980?

- A) Vaccinia virus
- B) Variola virus
- C) Cowpox virus
- D) Monkeypox virus

Q8. The two infectious forms of poxvirus particles are:

- A) Intracellular mature virus and extracellular enveloped virus
- B) Virion and nucleocapsid
- C) IMV and provirus
- D) EEV and defective interfering particle

Q9. Where does poxvirus DNA replication take place within the cell?

- A) In the nucleus
- B) In cytoplasmic viral factories, using virus-encoded DNA polymerase
- C) In mitochondria
- D) In the endoplasmic reticulum

Q10. The first human case of mpox occurred in 1970 in:

- A) United Kingdom
- B) Democratic Republic of the Congo
- C) China
- D) United States

Q11. Which clade of monkeypox virus has a case fatality rate of ~10.6%?

- A) Clade II (formerly West African)
- B) Clade I (formerly Congo Basin)**
- C) Clade IIb (2022 outbreak)
- D) Clade III

Q12. Tecovirimat acts by:

- A) Inhibiting viral DNA polymerase
- B) Inhibiting the VP37 membrane protein, preventing formation of enveloped virions needed for cell-to-cell spread**
- C) Blocking the M2 protein channel
- D) Inhibiting neuraminidase

Q13. Which antiviral drug is a lipid conjugate prodrug of cidofovir, FDA-approved for smallpox in June 2021, but with reported hepatotoxicity?

- A) Tecovirimat
- B) Brincidofovir**
- C) Cidofovir
- D) Acyclovir

Q14. The 3rd generation smallpox/mpox vaccine MVA-BN is:

- A) A live vaccinia virus vaccine with significant adverse events
- B) A highly attenuated, safer vaccine licensed for mpox prevention, and safe in immunocompromised patients and pregnancy**
- C) A formalin-inactivated whole-virion vaccine
- D) A DNA vaccine

Q15. Which of the following is a contraindication to vaccinia-based (1st/2nd generation) smallpox vaccines?

- A) Age over 50 years
- B) Immunocompromised patients, pregnancy, and eczema**
- C) A history of smallpox vaccination
- D) Allergy to eggs only

Lecture: Herpes viruses

Q1. How many human herpesviruses exist, and into how many subfamilies are they divided?

- A) 9 viruses in 3 subfamilies (alpha, beta, gamma)**
- B) 7 viruses in 2 subfamilies
- C) 12 viruses in 4 subfamilies
- D) 5 viruses in 1 subfamily

Q2. The herpesvirus particle is composed of 4 components. Which list is correct?

- A) Genome, capsid, tegument, and envelope with glycoprotein spicules**
- B) Genome, envelope, matrix, and reverse transcriptase
- C) Capsid, tail fibers, baseplate, and envelope
- D) Nucleocapsid, polymerase, envelope, and ribosome

Q3. During latency, the herpesvirus genome persists in the nucleus as:

- A) An integrated provirus

B) A circular DNA episome — NOT integrated into the cellular genome

C) A linear DNA fragment

D) An RNA replicon

Q4. Which subfamily of herpesviruses is described as leucotropic (infecting monocytes/macrophages) and causing infection in immunodeficiency and pregnancy?

A) Alphaherpesvirinae

B) Betaherpesvirinae (CMV, HHV-6/7)

C) Gammaherpesvirinae

D) Deltaherpesvirinae

Q5. Which diagnostic method is considered the gold standard for herpesvirus detection?

A) Serology IgM alone

B) Real-time PCR with quantification of viral load

C) Tzanck smear

D) MNI test

Q6. Epstein-Barr virus is transmitted mainly via saliva and shows dual cell tropism involving:

A) T lymphocytes and macrophages

B) B lymphocytes (latency) and epithelial cells (replication)

C) Natural killer cells and hepatocytes

D) Sensory neurons and epithelial cells

Q7. In the EBV serology table, the profile VCA IgM+, VCA IgG–, EBNA IgG– indicates:

A) Non-infected subject

B) Past infection

C) Primary infection / acute infectious mononucleosis

D) Reactivation

Q8. The MNI (Paul–Bunnell–Davidson) heterophile antibody test:

A) Is 100% specific for EBV

B) Has a false negative rate of 20% and false positives with HIV/CMV primary infection — not sufficient alone for diagnosis

C) Detects EBV DNA directly

D) Is the gold standard for diagnosis

Q9. Which cancer is 100% EBV-associated among HIV-infected patients?

A) Nasopharyngeal carcinoma

B) Cerebral lymphoma

C) African Burkitt lymphoma

D) Hodgkin lymphoma

Q10. For HSV-1 and HSV-2, the estimated global prevalence figures are:

A) HSV-1: ~3.7 billion people under age 50 (67%); HSV-2: 491 million people aged 15–49 (13%)

B) HSV-1: 13%; HSV-2: 67%

C) HSV-1: 491 million; HSV-2: 3.7 billion

D) HSV-1 and HSV-2: both ~1 billion

Q11. Herpes simplex meningoencephalitis is an emergency because:

- A) It is fatal without treatment in 80% of cases; antiviral treatment must start as soon as possible without waiting for results
- B) It always resolves spontaneously
- C) It only affects children under 2 years
- D) It cannot be diagnosed by CSF PCR

Q12. The Herpesviridae family is divided into three subfamilies based on their biological characteristics and genetic structures. Which of the following is NOT one of these subfamilies?

- A) Alphaherpesvirinae
- B) Betaherpesvirinae
- C) Deltaherpesvirinae
- D) Gammaherpesvirinae

Q13. For neonatal herpes prevention, which measure is recommended?

- A) Caesarean section decreases transmission if performed < 4 h after membrane rupture; maternal valaciclovir prophylaxis 1 g/day from 36 weeks
- B) Vaccination of the newborn at birth
- C) Immediate breastfeeding cessation in all cases
- D) Routine acyclovir for all pregnant women

Q14. A newborn suspected of HSV infection should receive:

- A) Oral aciclovir 400 mg ×3/day for 5 days
- B) Emergency IV aciclovir high dose 20 mg/kg/8h for 2–3 weeks, then oral
- C) Cidofovir IV once
- D) Paracetamol only

Q15. Which statement correctly describes varicella-zoster virus recurrence?

- A) Zoster is a reactivation in a neuron that travels back to the skin along the affected sensory nerve, with eruption limited to one dermatome segment
- B) Zoster is a generalized rash on the trunk
- C) Zoster always follows a second primary infection
- D) Zoster is caused by a different virus from varicella

Q16. What virus is one of the major opportunistic pathogens?

- A) Herpesviruses
- B) Cytomegalovirus
- C) Epstein-Barr Virus
- D) Varicella-Zoster Virus

Q17. Herpesviruses are characterized by their genetic material. What type of genome do all herpesviruses possess?

- A) Single-stranded RNA
- B) Double-stranded RNA
- C) Single-stranded DNA
- D) Double-stranded DNA

Q18. Which herpesvirus is responsible for causing both chickenpox (varicella) and shingles (herpes zoster)?

- A) Epstein-Barr Virus
- B) Cytomegalovirus

C) Varicella-Zoster Virus

D) Herpes Simplex Virus type 2

Q19. Herpes infection cause severe symptoms in which patient population?

A) Healthy adults

B) Immunocompetent patients

C) Individuals with strong immune responses

D) Neonates

Q20. Which is the direct diagnostic for Cytomegalovirus infection?

A) Ag detection

B) Virus isolation

C) Serology test

D) Real-time PCR

Lecture: Viral hepatitis

Q1. Which hepatitis virus is a "satellite virus" requiring HBsAg for packaging, release, and transmission?

A) HAV

B) HBV

C) HDV

D) HEV

Q2. Globally, the prevalence of HBsAg carriage is ~3.2%, corresponding to how many people living with chronic HBV?

A) ~257.5 million

B) ~71 million

C) ~62–72 million

D) ~820,000

Q3. In Cambodia (2022 survey), the HBsAg prevalence in the general population is:

A) 0.56%

B) 3%

C) 10.11%

D) 19.87%

Q4. Which group in Cambodia has the highest HBsAg prevalence (~19.87%)?

A) Children 5–7 years

B) Pregnant women

C) High-risk / co-infected populations

D) Children of HBsAg-positive mothers

Q5. The HBV genome is:

A) Linear double-stranded DNA of 125–230 kbp

B) Circular, partially double-stranded DNA of ~3,200 bp

C) Positive-sense RNA of 9.6 kb

D) Negative-sense circular RNA of 1,672–1,697 nt

Q6. Why is HBV difficult to cure?

- A) It integrates into the host genome
- B) cccDNA persists in the nucleus — the key to viral persistence, reactivation after antiviral therapy, and resistance to therapy**
- C) It replicates in the cytoplasm
- D) It mutates its surface antigen every year

Q7. HBeAg positivity indicates:

- A) Past infection with immunity
- B) Active viral replication**
- C) Vaccination
- D) Fulminant hepatitis only

Q8. A protective anti-HBs level is defined as:

- A) > 1 IU/L
- B) > 10 IU/L**
- C) > 100 IU/L
- D) Any detectable level

Q9. What percentage of adults clear HBV after acute infection, and what is the risk of chronicity?

- A) 90–95% clear; 5–10% progress to chronic infection**
- B) 50% clear; 50% chronic
- C) 75% clear; 25% chronic
- D) 35% clear; 65% chronic

Q10. Hepatitis D virus can only cause infection in individuals who are also infected with which other hepatitis virus?

- A) Hepatitis A virus
- B) Hepatitis B virus**
- C) Hepatitis C virus
- D) Hepatitis E virus

Q11. In Cambodia, which HCV genotypes are predominant?

- A) Genotype 3 and 4
- B) Genotype 1 (1b) and 6 (6e)**
- C) Genotype 2 only
- D) Genotype 5

Q12. Approximately what proportion of HCV infections progress to chronic infection?

- A) 5–10%
- B) ~35%
- C) ~75%**
- D) 90–95%

Q13. What is the treatment endpoint for HCV therapy with direct-acting antivirals?

- A) Undetectable HCV RNA (LLD $\leq$ 15 IU/mL) at SVR12 and/or SVR24 after end of treatment**
- B) Normalization of ALT only
- C) Negative anti-HCV antibody test
- D) Disappearance of HCV core antigen for 1 week

Q14. Which hepatitis viruses are transmitted by the faecal–oral route?

- A) HBV and HCV
- B) HAV and HEV**
- C) HDV and HEV
- D) HAV and HBV

Q15. According to the Cambodian data, HEV IgM positivity was found in what proportion of subjects with acute febrile illness in South-Central Cambodia?

- A) 5.5%
- B) 11.1%**
- C) 31.5%
- D) 20%

Q16. Which hepatitis viruses are known to commonly lead to chronic infection, increasing the risk of liver cirrhosis and hepatocellular carcinoma?

- A) Hepatitis A and E
- B) Hepatitis B and D
- C) Hepatitis B and C**
- D) Hepatitis A and D

Q17. Transmission of Hepatitis B, C, and D viruses primarily occurs through which of the following routes?

- A) Contaminated food and water
- B) Fecal-oral route
- C) Sexual contact and exposure to infected blood or bodily fluids**
- D) Airborne droplets

Q18. Which of the following is a common clinical manifestation of acute viral hepatitis, often observed as a yellowish discoloration of the skin and eyes?

- A) Rash
- B) Jaundice**
- C) Joint pain
- D) Diarrhea

Q19. Which hepatitis viruses have available vaccines for prevention?

- A) Hepatitis C and D
- B) Hepatitis A, B, and E**
- C) Hepatitis A and C
- D) Hepatitis B and E

Q20. Which diagnostic method is commonly used to identify specific viral markers and antibodies for different types of viral hepatitis?

- A) X-ray imaging
- B) Serological tests**
- C) Endoscopy
- D) Biopsy

Q21. Which of the following is a crucial prevention strategy for Hepatitis A and E?

- A) Safe injection practices
- B) Vaccination against HBV**

C) Ensuring adequate sanitation and food safety

D) Antiviral medication

Q22. What is a long-term complication associated with chronic Hepatitis B and C infections?

A) Pancreatitis

B) Kidney failure

C) Liver cirrhosis and cancer

D) Gallstones

Q23. What is the direct diagnosis for Hepatitis B screening?

A) Antigen and viral genome

B) IgM and IgG

C) Antigen and IgG

D) IgA and IgM